



AMERICAN INTERNATIONAL UNIVERSITY – BANGLADESH (AIUB)

Faculty of Engineering

Bachelor of Science in Electrical and Electronic Engineering/ Computer Engineering/ Computer Science & Engineering

EEE 2103 Electronic Devices (Section: All)

Final Exam.

Fall 2024-25

Total Marks: 40

Time: 2 hrs

Course Instructors: Dr. Md. Kabiruzzaman, Dr. Mohammad Hasan Imam, Dr. M. Tanzeer Ali, Dr. Effat Jahan, Dr. Md. Rifat Hazari, Dr. Nadia Afroze, Dr. Mohammad Alif Arman, Dr. Ebad Zahir, Ms. Susmita Ghosh, Ms. Bismoy Jahan, Mr. Abu Hena Md. Shatil, Ms. Sadia Yasmin, Mr. Ali Noor, Ms. Lia Moni, Mr. S M Tanvir Hassan Shoven, Mr. Md. Ashiquezzaman, Mr. Joheb Muhammad Tanzeer Sayeed, Mr. Shaoran Sayem, Mr. Md. Salehin Ferdous Kader, Ms. Samiha Ishrat Islam.

Instructions:

- Answer any **FOUR** from **Question 1** to **Question 5**.
- All the Answers should be written in the **ANSWER SCRIPT**.
- Questions of 2(a) and 4(b) correspond to CO3, others are aligned with CO4.
- Please return the question paper with the answer script.

1. (a) For the network given in Figure 1(a), analyze A_v by determining the values Z_i and Z_o for $R_D = 2 \text{ k}\Omega$ and $2.2 \text{ k}\Omega$ into the circuit. Given, $y_{fs} = 60 \text{ uS}$ and $g_m = 4 \text{ mS}$. [5]
- (b) For network of Figure 1(b), compare Q-point (V_{GSQ} , I_{DQ}) obtained from the mathematical [5] and graphical approach. Consider $I_{DSS} = 10 \text{ mA}$ and $V_P = -5 \text{ V}$. Explain the consistency of the results based on the principles of JFET operation and Q-point determination.

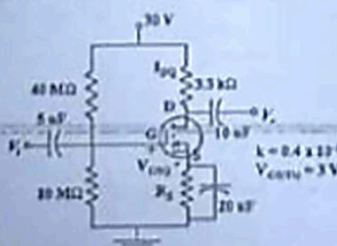


Figure for Question 1(a)

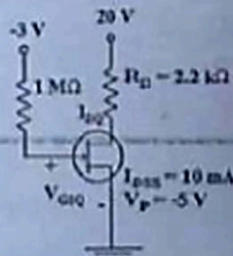


Figure for Question 1(b)

2. (a) For the network of following Figure 2(a), compare the values of A_v (with and without) [5] bypassed capacitor. Explain how the presence or absence of the bypass capacitor influences the gain, referencing the principles of transistor operation in AC.

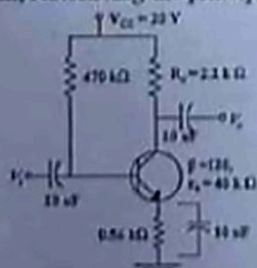


Figure for Question 2(a)

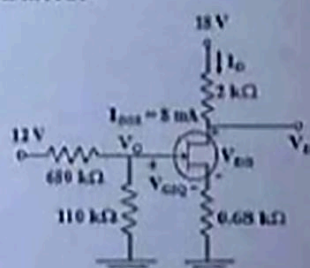


Figure for Question 2(b)

- (b) For given circuit in Figure 2(b), analyze I_D , V_{DS} , V_P for the values of $V_D = 12 \text{ V}$. [5]

SET: A

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Instructions:

• Answer any FOUR from Question 1 to Question 5.

• **All the Answers should be written in the ANSWER SCRIPT.**

• Questions of 2(a) and 4(b) correspond to CO3, others are aligned with CO4.

• *Please return the question paper with the answer script.*

1. (a) For the network given in Figure 1(a), analyze A, by determining the values Z_i and Z_o for R_p [5]

-2 k Ω and 2.2 k Ω into the circuit. Given, $\beta = 60$ and $r_e = 4 \text{ m}\Omega$.

(b) For network of Figure 1(b), compare Q-point (V_{CE}) obtained from the mathematical [5] and graphical approach.

Consider $I_{BSS} = 10 \text{ mA}$ and $V_{CC} = 5 \text{ V}$. Explain the consistency of the results based on the principles of JFET operation and Q-point determination.

40 $\text{M}\Omega$

MA

Figure for Question 1(a)

k-84x13 Very-3V

20 V

-3V

IM Ω

Rp-2.2 k

$I_{BSS} = 10 \text{ mA}$

Figure for Question 1(b)

2. (a) For the network of following Figure 2(a), compare the values of A_v (with and without) [5] bypassed capacitor. Explain how the presence or absence of the bypass capacitor influences the gain, referencing the principles of transistor operation in AC.

18 V

3749

47044

12V

74410

680

110 kn

05410

14x7

ش

1116

6.68 kn

Figure for Question 2(a)

Figure for Question 2(b)

(b) For given circuit in Figure 2(b), analyze I_n , V_{as} , V_r for the values of $V_p = 12$ V.

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[S]